Joint Generalized Lq and Convolutional Regularization: Enhancing mmW Automotive SAR Sparse Imaging

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***Abstract*— Millimeter-wave (mmW) automotive synthetic aperture radar (Auto-SAR) technology holds significant promise for advanced driver assistance systems (ADASs). Sparse imaging methods can improve the quality of Auto-SAR images, such as suppressing sidelobes and noise. However, the l_1 convex regularization-based sparse imaging methods suffer from the bias estimation, which reduces the target amplitude and ignores the association between scatterers, weakening the target structure. To address these issues, we proposed joint generalized l_q and convolutional (Glq-Con) regularization to enhance mmW Auto-SAR sparse imaging in this article. First, to improve the target amplitude, we propose utilizing the nonconvexity of Glq to reduce the bias effect; meanwhile, the global convergence of Glq ensures the imaging accuracy. Then, considering the continuity of the imaging target in driving scenes, we propose to utilize convolution regularization to modify the previously reconstructed amplitude of Glq to improve the target structure. Besides, to reduce computational complexity, we establish an efficient sparse imaging model. In this model, the fast Fourier transform (FFT) operator is employed to approximate complex matrix operation in the iterative process. We also use an efficient optimizer to solve the imaging model. Finally, both simulations and measured typical driving scenario experiments demonstrate that the proposed method significantly enhanced the Auto-SAR image, especially for the targets of weak scatterers.**

***Index Terms*— Automotive radar, convolutional regularization, enhancing imaging, synthetic aperture radar (SAR), sparse imaging.**

I. Introduction

MILLIMETER-WAVE (mmW) automotive radar plays an important role in advanced driver assistance systems (ADASs), including adaptive cruise control (ACC), unseen areas detection (UAD), and lane change assist (LCA) [1], [2], [3]. Compared to other sensors, e.g., camera and lidar, it still provides reliable environmental information for ADAS, even in adverse weather conditions e.g., fog, rain, and snow, through radar imaging technology [4]. The automotive radar can achieve centimeter-level range resolution by utilizing 77-GHz frequency-modulated continuous wave (FMCW) with large sweep bandwidth [5]. Unfortunately, the antenna array aperture length is limited by cost and size in the current commercial FMCW radar, which results in lower azimuth/angular resolution. Therefore, how to improve the azimuth resolution and imaging quality of FMCW automotive radar has become a current research hotspot in academia and industry.

Synthetic aperture radar (SAR) technology has gradually emerged in the automotive field in recent years [6], [7], [8], [9], [10], [11], [12], [13], [14], [15]. It can achieve high-resolution imaging by the platform movement to form a large virtual aperture to improve the azimuth/angular resolution, which does not need to pay any additional hardware cost [16]. In [7], [8], [9], and [10], an automotive SAR (Auto-SAR) imaging system is designed to achieve a high-resolution image by using the conventional matched filtering (MF) imaging method [17]. In [15], a polarimetric Auto-SAR system has been designed to obtain high-resolution images of various urban scenes by the conventional range migration algorithm (RMA) imaging method [17]. However, these imaging methods can result in high sidelobes and noise interference, which will significantly impact the detection and recognition performance of the ADAS in the driving scene.

Recently, sparse imaging methods based on compressed sensing (CS) theory have attracted widespread attention in radar signal processing [18], [19], [20], [21], [22], [23], [24], [25], [26]. It primarily utilizes radar echo data and the measurement matrix of the system to establish a linear measurement model and address ill-posed linear inverse problems by incorporating a sparse regularization prior. In particular, sparse regularization plays a key role in reducing high sidelobes, suppressing background noise, and improving imaging resolution [27]. Commonly, the l_1-norm as convex sparse regularization is widely used in sparse imaging, which is formulated into the least absolute shrinkage and selection operator (LASSO) problem [28]. To solve this problem, many threshold function shrinkage algorithms have been proposed, such as the iterative shrinkage-thresholding algorithm (ISTA) [29], and the approximate message passing (AMP) algorithm [30]. However, the l_1-norm is relaxed by the l_0-norm, resulting in the bias problem by the fixed soft shrink-

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age threshold, which reduces the reconstructed amplitude [31]. In imaging results, the bias problem leads to a lack of target original structure contours when applied to sparse imaging, which will seriously affect the imaging quality. In particular, there are usually weak RCS targets in vehicle radar imaging scenes, e.g., pedestrians and tree lawns.

To reduce the bias effect and improve the reconstruction accuracy, we focus on the nonconvex regularization. Compared to the l_1-norm, the optimization properties of the nonconvex regularization are closer to the l_0-norm [32]. Therefore, these optimization algorithms based on nonconvex regularization are considered to have the potential to improve bias effects and enhance the performance of reconstruction accuracy. Various algorithms have been developed using nonconvex regularization, popular examples of them include: the algorithms based on threshold function, e.g., smoothly clipped absolute deviation (SCAD) [33] and minima concave penalty (MCP) [34]. Typically, these algorithms need to adjust more hyperparameters than ISTA. The algorithms based on l_q-norm with $0 < q < 1$ are processed iteratively, typical algorithms [35], [36]. In particular, comparing other nonconvex algorithms, these iteration-based l_q-norm algorithms have fewer hyperparameters and simpler expression. However, the matrix inversion operations present in the iteration-based l_q-norm algorithm exacerbate the computational complexity. Besides, these regularizations have difficulty converging to the global optimum. Based on the above situation, it drives us to seek a regularization that is more suitable for Auto-SAR sparse imaging. Fortunately, Candes et al. [37] first proposed a generalized l_q-norm regularization for sparse coding-based image restoration, which took advantage of the idea of threshold function. It mainly realizes the convergence of the global optimal solution by designing an adaptive threshold. It inspired us to propose the use of generalized l_q regularization (Glq) to improve the imaging accuracy in Auto-SAR sparse imaging.

On the other hand, the sparse imaging methods based on sparse regularization effectively suppress sidelobes and background noise. However, these methods ignore the association between scatterers of targets, especially the imaging targets in automotive radar imaging applications, such as cars, electric vehicles, and signs, which usually have a structure of continuity. These methods that lead to Auto-SAR images are often composed of several strong scatterers that are separated from each other, and their visual quality is actually limited. When these Auto-SAR images are applied in the ADAS system, it will be difficult to extract the shape contour and mechanism features of the target. It inspires us to establish the connection between the target scatterers to enhance the target structure.

Besides, the large measurement matrix of sparse imaging usually has high computational complexity and storage space. In our previous work, the measurement matrix is further replaced by the smaller spatial Fourier matrix [38]. Meanwhile, sparse imaging methods based on the idea of approximate observations have been widely researched in SAR remote sensing applications [39]. Its basic principle is to replace the measurement matrix with the traditional imaging process, which avoids substantial matrix operations and reduces the computational complexity. It inspires us to employ the fast Fourier transform (FFT) operator to replace the matrix operation to reduce the computational complexity.

In this work, we dedicate ourselves to the problem of mmW automotive radar imaging and joint generalized and convolutional (Glq-Con) regularization for sparse imaging to enhance mmW Auto-SAR image quality. First, comparing with l_1 regularization, we propose utilizing the nonconvexity of Glq to reduce the bias effect to improve the estimated amplitude of the imaging target. Then, we consider the continuity of imaging target in driving scenes and propose a convolution regularization to improve the target structure. Next, we also propose employing the FFT operator to approximate the matrix operation to computational complexity. Finally, both numerical and visual results demonstrate proposed method yields compelling reconstruction performance while maintaining efficient computing performance.

The main contributions of this article are as follows.

1) We proposed joint Glq-Con regularization to enhance Auto-SAR imaging. More precisely, a Glq is proposed for improving the target amplitude. It effectively reduces the bias problem of l_1 regularization and can converge efficiently to the global optimum. A Con regularization is proposed for improving the target structure. It modifies the reconstructed amplitude of Glq by using the continuity of imaging targets in driving scenes.
2) We proposed to employ the FFT operator to establish an efficient sparse imaging model with Glq-Con regularization. In this model, the FFT operator approximates complex matrix operation in the iterative process. It avoids large measurement matrix operations, effectively reducing the time complexity. We also use an efficient optimizer to solve the imaging model.
3) Simulations and typical driving scenario experiments are conducted. The results show that the proposed Glq-Con sparse imaging method achieves compelling imaging performance while maintaining high computational speed.

The rest of this article is organized as follows. Section II introduces the Auto-SAR sparse imaging model. Section III introduces our methods. The results are shown in Section IV. Finally, Section V summarizes this entire article.

II. Auto-SAR Sparse Imaging Model

The imaging scene of Auto-SAR system is shown in Fig. 1. The radar transmits an FMCW signal, which is expressed as

$$s_t(t) = \exp\left\{ j2\pi \left(f_0 t + 0.5\gamma t^2 \right) \right\}, \quad |t| \le T_s/2 \tag{1}$$

where f_0 is the carrier frequency; $\gamma = B/T_s$ is the modulation rate; and B and T_s are the sweep bandwidth and pulse duration, respectively. $t = 0, \ldots, N$ is the radar fast time and N is the number of the fast time sampling.

In the Auto-SAR imaging scene, the vehicle platform is moving at v_s, and the Auto-SAR is configured side-looking. When the mth pulse is transmitted, the position of the radar is (x_m, y_m). For the whole imaging scene, the echo is obtained by adding the received signals of the imaging scene $\sigma(x_a, y_b)$,

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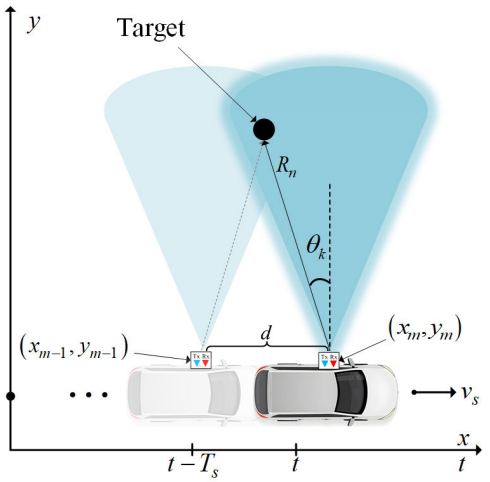


Fig. 1. Auto-SAR system and imaging scene: the vehicle is moving horizontally at velocity v_s, and radar side looking the target.

and the echo data of the Auto-SAR are expressed as

$$s(n,m)=\sum_{a=1}^{N_1}\sum_{b=1}^{Q}\sigma(x_a,y_b)\cdot\exp\{-j4\pi\gamma nT_sR_m(x_a,y_b)/c\}\times\exp\{-j4\pi R_m(x_a,y_b)/\lambda\} \tag{2}$$

where x_a and y_b represent the azimuth and range directions concerning the imaging scene σ, respectively; and N_1 and Q represent the number of targets in the range and azimuth directions, respectively. The range from the target to the radar at the mth pulse is $R_m(x,y)=((x_m-x_a)^2+(y_m-y_b)^2)^{1/2}$. $m=1,\ldots,M$ is the azimuth sampling and M is the number of SAR channels. $\lambda=f_0/c$ is the wavelength and c is the speed of light.

After range FFT and range migration correction (RMC) in (2), the echo data are further represented as

$$\hat{y}(\eta,m)=\mathcal{F}\{s(n,m)\}=\sum_{\eta=1}^{N}\sum_{b=1}^{Q}\delta_\eta(x_a,y_b)\cdot\exp\{-j4\pi R_m(x_a,y_b)/\lambda\} \tag{3}$$

where $\mathcal{F}\{\cdot\}$ represents the FFT operation, and the scattering coefficient of the point scatterer after range FFT is given by $\delta_\eta(x_a,x_b)=\sigma(x_a,x_b)\cdot\mathrm{sinc}\{\gamma T(\eta-2R_m(x_a,x_b)/c)\}$, in which $\eta=1,\ldots,N$ is represented as the ηth range unit corresponding to the nth fast time sampling and $sinc\{\cdot\}$ represents the sampling function.

In (3), there is a coupling between the azimuth and range directions, which requires a large measurement matrix to characterize the relationship. In our previous work, the azimuth and range directions are decoupled by an effective phase compensation process [38]. Then, the echo data are further represented as

$$y(\eta,m)=\sum_{\eta=1}^{N}\sum_{k=1}^{Q}\delta(\eta,\theta_k)\cdot4\pi md\sin\theta_k/\lambda \tag{4}$$

where $k=1,\ldots,Q$, θ_k is the angle of target, and d is the distance of SAR channels.

Then, the Auto-SAR sparse imaging model can be expressed as

$$\mathbf{Y}=\mathbf{AX}+\tilde{\mathbf{N}}\in\mathbb{C}^{M\times N} \tag{5}$$

where $\mathbf{Y}=[\mathbf{y}_1,\ldots,\mathbf{y}_N]\in\mathbf{C}^{M\times N}$ and $\mathbf{X}\in\mathbb{C}^{Q\times N}$ are the collected 2-D echo data and 2-D imaging scene, respectively; $\mathbf{y}_n=[y(n,1),\ldots,y(n,M)]^T\in\mathbb{C}^M$, and $\delta(\eta,\theta_k)$ is the element of $\mathbf{X}$. $\mathbf{A}=[\mathbf{a}(\theta_1),\ldots,\mathbf{a}(\theta_Q)]\in\mathbb{C}^{M\times Q}$ is the measurement matrix, and $\mathbf{a}(\theta_k)=[0,\ldots,e^{-j4\pi(M-1)d\sin\theta_k/\lambda}]^T$. $\tilde{\mathbf{N}}\in\mathbb{C}^{M\times N}$ is the noise matrix.

In sparse imaging methods, the l_1-norm, as the regularization, can be efficiently solved and obtain a sparse solution to the considered problem. According to the model of (5), the usual regularization technique has the following form:

$$\min_{\mathbf{X}}\frac{1}{2}\|\mathbf{Y}-\mathbf{AX}\|_2^2+\beta(\mathbf{X};\lambda) \tag{6}$$

where β is the regularization function and λ is the regularization parameter. In (6), the l_1-norm is usually used as the regularization function to efficiently solve and obtain a sparse solution to the considered problem. Thus, the Auto-SAR sparse imaging model based on l_1-norm is as

$$\min_{\mathbf{X}}\frac{1}{2}\|\mathbf{Y}-\mathbf{AX}\|_2^2+\lambda\|\mathbf{X}\|_1. \tag{7}$$

In (7), l_1-norm is a convex relaxation constraint with respect to l_0-norm, which results in a biased estimation problem in the amplitude of target. Therefore, l_1 regularization cannot meet the demand for higher accuracy imaging due to the bias estimate.

Besides, the measurement matrix usually incurs expensive computational costs when solving the above model. Based on the abovementioned problems, our work can be carried out in calculation load and imaging accuracy.

III. Methodology

In this section, it consists of four parts. First, we analyze the biased estimation of l_1 regularization. Then, to improve the biased estimation, we introduce the Glq regularization and its global convergence. Next, to further improve the imaging quality, we introduce convolutional regularization by using the continuity of imaging targets in driving scenes. Finally, to improve the imaging efficiency, we introduce an FFT operator-based efficient sparse imaging model with Glq-Con regularization, and we also use an efficient optimizer to solve the imaging model.

A. Analysis of Bias Estimation of l_1 Regularization

Generally, the corresponding threshold function is used to solve the problem with different regularization in (6) [32], Therefore, from the perspective of the threshold function, we can analyze the bias estimation of l_1-norm regularization in detail.

First, for the l_1-norm regularization in (7), the threshold function is designed to be the soft thresholding function [29]

$$\bar{\mathbf{X}}=\mathrm{soft}(\mathbf{X};\lambda)=\begin{cases}0, & |\mathbf{X}|\le\lambda\\ (|\mathbf{X}|-\lambda)\mathrm{sign}(\mathbf{X}), & |\mathbf{X}|>\lambda\end{cases} \tag{8}$$

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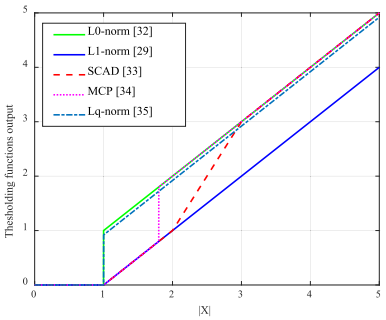


Fig. 2. Curves of thresholding functions of l_0-norm, l_1-norm, and typical nonconvex regularization when $\mathbf{X}$ is a 1×1 matrix (with the same threshold).

where $\bar{\mathbf{X}}$ is the output of the soft thresholding function and sign$(\cdot)$ is the sign function. When the measurement matrix $\mathbf{A}$ is determined, the soft thresholding function is the main factor that influences the bias of the result.

Then, consider that the case of nonzero values of $\mathbf{X}$ is greater than λ in soft$(\mathbf{X}; \lambda)$ and according to [41]. Therefore, the bias estimation of l_1-norm regularization can be described as

$$\begin{aligned}\bar{\mathbf{X}} &= \left(\mathbf{A}^H\mathbf{A}\right)^{-1}\left\{\mathbf{X} - \lambda \text{sign}(\mathbf{X})\right\} \\ &= \left(\mathbf{A}^H\mathbf{A}\right)^{-1}\mathbf{A}^H\mathbf{Y} - \lambda\left(\mathbf{A}^H\mathbf{A}\right)^{-1}\text{sign}(\mathbf{X})\end{aligned} \tag{9}$$

where the first term of (9) equals the oracle estimator $\bar{\mathbf{X}}$, which is unbiased, whereas the second term of (8) never vanishes and is the bias. Therefore, according to (9), the solution of (7) based on l_1 regularization is the biased estimation according to statistical signal processing.

In addition, we further illustrate the bias problem by performing a numerical analysis of the threshold function in Fig. 2. In the numerical analysis, the soft threshold function of l_1 regularization [29] is compared with some nonconvex threshold functions of l_0-norm [32], SCAD [33], MCP [34], and l_q-norm with $0 < q < 1$ [35], and these nonconvex threshold functions have been researched for reducing bias estimates. In Fig. 2, it can be seen that the amplitude results of SCAD, MCP, and l_q-norm are closer to l_0-norm, but the amplitude of l_1-norm is severely reduced than l_q-norm. Therefore, nonconvex regularization can effectively improve the bias problem. However, these nonconvex threshold functions cannot converge to the global optimal solution, because of their nonconvex properties [37]. This has the potential to reduce the accuracy of the imaging. Next, we will introduce a Glq nonconvex regularization to further improving the accuracy.

B. Generalized Lq Nonconvex Regularization

In the above nonconvex regularization, they mainly use a threshold function to perform segmentation to obtain the optimal solution. Based on this principle, we can modify the threshold and contraction rules to make the l_q-norm converge to the global optimal solution, and details are as follows.

First, considering the l_q-norm regularization in (6), then, let x and y be elements of matrices $\mathbf{X}$ and $\mathbf{Y}$, respectively. Without loss of generality, in the following, we only consider the case of $y > 0$ and it is differentiable in the range of $(0, +\infty)$. Then, the l_q-norm problem is given as

$$f(x) = \frac{1}{2}(x - y)^2 + \lambda|x|^q \tag{10}$$

where $0 < q < 1$, and when $q = 1$, it is an l_1-norm problem.

Then, we can easily get the results for the first and second derivatives of $f(x)$

$$f'(x) = x - y + \lambda q x^{q-1} \tag{11}$$

$$f''(x) = 1 + \lambda q(q-1)x^{q-2} \tag{12}$$

Let f'', and we have

$$x_0 = \left\{\lambda q(1-q)\right\}^{\frac{1}{2-q}}. \tag{13}$$

It can be easily verified that $f(x)$ is concave in the range of $(0, x_0)$ and is convex in the range of $(x_0, +\infty)$ too. Therefore, we can set a generalized soft thresholding $\beta^{\text{Glq}}(\lambda)$ to make the original nonconvex problem transform two concave and convex problems. When $y < \beta^{\text{Glq}}(\lambda)$, $x = 0$ is the global minimum; otherwise, the nonzero solution would be optimal. Therefore, according to the Appendix, we should solve the following nonlinear equation system to determine a correct thresholding value $\beta^{\text{Glq}}(\lambda)$ and its corresponding x^*:

$$\begin{cases}\left\{x^* - \beta^{\text{Glq}}(\lambda)\right\}^2 + 2\lambda\left(x^*\right)^q = \left\{\beta^{\text{Glq}}(\lambda)\right\}^2 \\ x^* - \beta^{\text{Glq}}(\lambda) + \lambda q\left(x^*\right)^{q-1} = 0.\end{cases} \tag{14}$$

Then, we can obtain the following equation:

$$\left(x^*\right)^q\left\{2\lambda(1-q) - \left(x^*\right)^{2-q}\right\} = 0. \tag{15}$$

Thus, the only solution of x^* in the range of $(x_0, +\infty)$ can be given as

$$x^* = \left\{2\lambda(1-q)\right\}^{\frac{1}{2-q}}. \tag{16}$$

We substituted (15) into (13) and can obtain the thresholding

$$\beta^{\text{Glq}}(\lambda) = \left\{2\lambda(1-q)\right\}^{\frac{1}{2-q}} + \lambda q\left\{2\lambda(1-q)\right\}^{\frac{q-1}{2-q}}. \tag{17}$$

When $y > \beta^{\text{Glq}}(\lambda)$, we can obtain the global optimal solution by the following equation in $(x^*, +\infty)$:

$$\beta^{\text{Glq}}(y;\lambda) - y + \lambda q\left\{\beta^{\text{Glq}}(y;\lambda)\right\}^{q-1} = 0. \tag{18}$$

Consider the whole imaging scenario $\mathbf{X}$ and the threshold function $\beta^{\text{Glq}}(\mathbf{X}; \lambda)$, and the Glq regularization-based sparse imaging model can be represented as

$$\min_{\mathbf{X}} \frac{1}{2}\|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_2^2 + \beta^{\text{Glq}}(\mathbf{X};\lambda). \tag{19}$$

In this model, we can get the global optimal solution of l_q regularization by threshold function $\beta^{\text{Glq}}(\mathbf{X}; \lambda)$, i.e., when $\mathbf{Y} < \beta^{\text{Glq}}(\lambda)$, $\mathbf{X} = 0$, and when $\mathbf{Y} > \beta^{\text{Glq}}(\lambda)$, the global minimum $\mathbf{X}$ can be obtained by solving (17). The thresholding function is specifically described in Algorithm 1.

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Algorithm 1 Glq Thresholding Function

Input: $q, \lambda, \mathbf{X}$

$\tau^{Glq} = \{2\lambda(1-q)\}^{1/(2-q)} + \lambda q\{2\lambda(1-q)\}^{(q-1)/(2-q)}$

if $|\mathbf{X}| \le \tau^{Glq}$

$\beta^{Glq}(\mathbf{X};\lambda) = 0$

else $j = 0, \mathbf{X}_{Glq}^{(j)} = |\mathbf{X}|$

Iterate on $j = 0, 1, 2$

$\mathbf{X}_{Glq}^{(j+1)} = |\mathbf{X}| - \lambda q\left(\mathbf{X}_{Glq}^{(j)}\right)^{q-1}$

$j = j + 1$

$\beta^{Glq}(\mathbf{X};\lambda) = \mathbf{sign}(\mathbf{X})\mathbf{X}_{Glq}^{(j+1)}$

end

Output: $\beta^{Glq}(\mathbf{X};\lambda)$

C. Joint Generalized Lq and Convolutional Regularization

In (18), the amplitude of the imaging target is effectively preserved, especially for some weak targets whose structure contour is lost because of the bias effect. From the imaging results, the traditional sparse regularization often assumes that radar images consist of isolated scatterers. However, in Auto-SAR imaging applications, imaging targets are typically a structure of continuity scatterers, such as cars, electric vehicles, and signs. The traditional sparse regularization ignores the correlation between the scatterers. It leads the obtained radar images to fail to achieve satisfactory visual effects, which affects the target detection and recognition tasks based on radar images.

In order to further improve the structural features of the imaging target, we consider the continuous scatterer structure, and we use the convolutional regularization to modify the previously reconstructed amplitude of Glq. The improved joint Glq-Con regularization sparse imaging model can be described as

$$\begin{cases} \min\limits_{\mathbf{X}} \dfrac{1}{2}\|\mathbf{Y} - \mathbf{A}\mathbf{X}\|_2^2 + \beta^{\mathrm{Glq}}(\boldsymbol{\omega} \odot \mathbf{X};\lambda) \\ \boldsymbol{\omega} = 1/\{\mathcal{C}(\mathbf{C},\mathbf{X}) + \varepsilon\} \end{cases} \tag{20}$$

where $\boldsymbol{\omega} \in \mathbb{C}^{Q\times N}$ is the convolutional regularization, $\odot$ is the Hadamard product, and ε is a small constant to avoid singularity. $\mathcal{C}(\cdot)$ is a convolution operator and $\mathbf{C}$ is the convolution kernel.

Convolution kernel $\mathbf{C}$ can effectively enhance Auto-SAR images. When the value of the neighboring points is close to the value of the scattering center point, the continuity structure of the target can be maintained. Therefore, we can set the convolution kernel $\mathbf{C}$ as follows:

$$\mathbf{C} = \begin{bmatrix} b & b & b \\ b & a & b \\ b & b & b \end{bmatrix} \tag{21}$$

where the center point a is set to 1 and the neighboring point $b \in [0.5, 1]$.

D. Efficiently Glq-Con Sparse Imaging Model

In (19), the main computational complexity comes mainly from matrix operations about the measurement matrix $\mathbf{A}$. Therefore, we use FFT operations to approximately replace the matrix operations of the measurement matrix to reduce the computation complexity. In (19), it can be supposed that there is an approximation relationship between the traditional imaging procedure and the complex conjugate multiplication of the measurement matrix $\mathbf{A}$ with the imaging scene matrix $\mathbf{X}$

$$\mathbf{Y} = \mathcal{I}^{-1}(\mathbf{X}) \cong \mathbf{A}\mathbf{X} \tag{22}$$

$$\mathbf{X} = \mathcal{I}(\mathbf{Y}) \cong \mathbf{A}^H\mathbf{Y} \tag{23}$$

where $(\cdot)^H$ and $(\cdot)^{-1}$ denote the complex conjugate transpose and matrix inversion operation, respectively, and $\mathcal{I}(\cdot)$ denotes the imaging operator.

In (21) and (22), $\mathbf{A}$ can be thought of as an overcomplete dictionary of spatial Fourier transforms matrix; thus, we can approximate the $\mathbf{A}$ matrix with FFT processing in the azimuth dimension

$$\mathcal{I}(\cdot) = \mathcal{F}(\cdot), \mathcal{I}^{-1}(\cdot) = \mathcal{F}^{-1}(\cdot) \tag{24}$$

where $\mathcal{F}(\cdot)$ and $\mathcal{F}^{-1}(\cdot)$ are the FFT operator and inverse FFT operator, respectively.

Therefore, we proposed an efficient sparse imaging model, which is expressed as

$$\min_{\mathbf{X}} \frac{1}{2}\left\|\mathbf{Y} - \mathcal{F}^{-1}(\mathbf{X})\right\|_2^2 + \beta^{\mathrm{Glq}}(\omega \odot \mathbf{X};\lambda). \tag{25}$$

In the above imaging model, we propose to employ an efficient optimizer to solve this problem, i.e., majorization–minimization (MM) ideal [41].

Let $g(\mathbf{X}) = \|\mathbf{Y} - \mathcal{F}^{-1}(\mathbf{X})\|_2^2$, and the establishing following surrogate function can be utilized to replace the objective function in (24):

$$\tilde{g}(\mathbf{X}, \mathbf{X}^{i-1}) = g(\mathbf{X}, \mathbf{X}^{i-1}) + \langle \mathbf{X} - \mathbf{X}^{i-1}, \nabla g(\mathbf{X}^{i-1})\rangle + \frac{1}{2\mu}\left\|\mathbf{X} - \mathbf{X}^{i-1}\right\|_2^2 + \beta^{\mathrm{Glq}}(\boldsymbol{\omega} \odot \mathbf{X};\lambda) \tag{26}$$

where μ is the proximal parameter and i is the ith iteration.

Also, ∇g is the gradient operator of $g(\mathbf{X})$, which is derived as

$$\nabla g(\mathbf{X}^{i-1}) = -\mathcal{F}\{\mathbf{Y} - \mathcal{F}^{-1}(\mathbf{X}^{i-1})\}. \tag{27}$$

For each iteration $\mathbf{X}^i$, it can obtain the following relationship to ensure the convergence of the iteration:

$$g(\mathbf{X}^i) \le \tilde{g}(\mathbf{X}^i, \mathbf{X}^{i-1}) \le \tilde{g}(\mathbf{X}^{i-1}, \mathbf{X}^{i-1}) = g(\mathbf{X}^{i-1}). \tag{28}$$

In (25), by ignoring the constant terms and merging the like terms, it can be further simplified as

$$\tilde{g}(\mathbf{X}^i, \mathbf{X}^{i-1}) = \frac{1}{2\mu}\left\|\mathbf{X}^i - (\mathbf{X}^{i-1} - \nabla g(\mathbf{X}^{i-1}))\right\|_2^2 + \beta^{\mathrm{Glq}}(\boldsymbol{\omega} \odot \mathbf{X}^{i-1};\lambda). \tag{29}$$

After the threshold function, the solution of (28) can be given as

$$\mathbf{X}^i = \beta^{\mathrm{Glq}}\left\{\mathbf{X}^{i-1} - \nabla g(\mathbf{X}^{i-1}); \frac{\mu\lambda}{\{\mathcal{C}(\mathbf{C}, \mathbf{X}^{i-1}) + \varepsilon\}}\right\}. \tag{30}$$

In this imaging model, we use the FFT operator to simplify the measurement matrix operations and the MM optimizer to

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simplify the process of solving the model. It offers a potential for real-time or semi-real-time Auto-SAR sparse imaging, and the detailed description can be seen in Algorithm 2.

Algorithm 2 Proposed Glq-Con Sparse Imaging

Input: q, λ, $\mathbf{Y}$, μ, $\boldsymbol{\omega}$, ε, maximum number of iterations I_{max}, and error parameter $\hat{\varepsilon}$

Initialize the variables $\tilde{\mathbf{X}}$, $\mathbf{X}^{(0)}$, $t^{(0)}$ as zeros, and set $i = 1$
for $i \le I_{max}$ **do**
  $t^{(i)} = 0.5\left\{1 + \sqrt{1 + 4(t^{(i-1)})^2}\right\}$
  $\mathbf{X}^{(i)} = \mathbf{X}^{(i-1)} + \frac{t^{(i-1)} - t^{(i)}}{t^{(i)}}\left(\mathbf{X}^{(i-1)} - \mathbf{X}^{(i-2)}\right)$
  $\tilde{\mathbf{X}} = \mathbf{X}^{(i)} + \mu\mathcal{F}\left\{\mathbf{Y} - \mathcal{F}^{-1}\left(\mathbf{X}^{(i)}\right)\right\}$
  $\mathbf{X}^{(i+1)} = \beta^{Glq}\left(\boldsymbol{\omega} \odot \tilde{\mathbf{X}}; \mu\lambda\right)$
  if $\left\|\mathbf{X}^{(i+1)} - \mathbf{X}^{(i)}\right\|_2^2 \Big/ \left\|\mathbf{X}^{(i)}\right\|_2^2 > \hat{\varepsilon}$ **do**
    $j = j + 1$
  else break
end

Output: imaging scene $\hat{\mathbf{X}} = \mathbf{X}^{(i+1)}$

IV. Results

In this section, we analyze the performance of the proposed sparse imaging method on mmW Auto-SAR imaging with both simulations and experiments (two driving scenarios by different target angles). The superiority of the proposed method is demonstrated by comparing with traditional MF and l_1 regularization-based ISTA. We conducted both quantitative and qualitative evaluations on imaging quality and computational complexity. In simulations, different SNR cases were also considered. First, we established a point target experiment to verify the bias estimation and the improved performance of Glq on the bias estimation. Then, we established a car target experiment and conducted a detailed analysis of the imaging results. Then, for the real experiments, we establish an Auto-SAR imaging platform with a TI AWR2243 EVM radar in Fig. 3. In this platform, the AWR2243 radar is configured with 1T1R mode, and a GNSS provides the supplementary platform velocity information. In the real experiments, there are two typical vehicular driving scenarios that are carried out, and the specific system simulation and real platform parameters are shown in Table I. In order to better evaluate the accuracy and performance of the mmW Auto-SAR imaging method, the normalized image entropy (NIE) [42], peak-to-background ratio (PBR) [43], and normalized average target region intensity (NAT) [43] are used in this article

$$\mathrm{NIE} = -\sum_{i=1}^{N}\sum_{j=1}^{Q}\frac{|x_{ij}|^2}{\|\bar{\mathbf{x}}\|_2^2}\log\frac{|x_{ij}|^2}{\|\bar{\mathbf{x}}\|_2^2}, \quad x_{ij} \in \bar{\mathbf{x}} \tag{31}$$

where $\bar{\mathbf{X}}$ is the normalized radar image, $\bar{\mathbf{X}} = |\hat{\mathbf{X}}/\max(\hat{\mathbf{X}})|$, $\hat{\mathbf{X}}$ is the imaging result, and x_{ij} is the pixel value of $\hat{\mathbf{X}}$

$$\mathrm{PBR} = 20\log_{10}\left(\frac{N_b g_t^{\mathrm{pk}}}{\bar{g}_b}\right) \tag{32}$$

where g_t^{pk} is the target peak in $\hat{\mathbf{X}}$, $\bar{g}_b = \sum_{i=1}^{N_b}|g_i|$, $g_i \in \mathbf{g}_b$ is represented as the sum of background pixels, and N_b is the total number of background pixels

$$\mathrm{NAT} = \frac{\bar{g}_t}{\bar{N}_t} \tag{33}$$

where $\bar{g}_t = \sum_{i=1}^{\bar{N}_t}|g_i|$, $g_i \in \mathbf{G}_t$ is the sum of pixels in the potential target region $\mathbf{G}_t$ and $\bar{N}_t$ is the total number of pixels in $\mathbf{G}_t$.

In these evaluation indexes, the larger PBR, the higher the background noise suppression capability, and vice versa. NAT is used to evaluate the performance of ISTA, Glq, and Glq-Con in target structure reconstruction. The higher NAT, the better the target structure reconstruction, and vice versa. The lower NIE, the better the imaging quality in different SNRs, and vice versa. Meanwhile, considering the performance of target structure enhancement in these sparse imaging methods in the same SNR, the larger NIE that the target structure is obvious and the imaging quality is better. In addition, all experiments are reported from personal computer (PC) tests on MATLAB 2020a with CPU i9-10900KF and GPU RTX 3070 with 64-GB RAM. In the simulations and experiments, $I_{\max}$ is set to 20, $\varepsilon = 10e^{-6}$, error parameter $\hat{\varepsilon} = 10e^{-3}$, and the size of the convolution kernel $\mathbf{C}$ is set to 3×3, $a = 1$. The proximal parameter $\mu = 0.5\|\mathbf{A}^H\mathbf{A}\|_2^2$ and the regularization parameter λ in (20) are initialized by the variance of the echo data $\mathbf{Y}$, as $\lambda = [1, 5] \times \sigma_{\mathbf{Y}}^2$. The azimuth (angle) imaging space is set to $\Delta\theta = [-30°, 30°]$. Meanwhile, in simulations, b is set to 0.5 and q is set to 0.7, and in experiments, b is set to 1 and q is set to 0.7.

TABLE I
Radar System Parameter

Parameter	Simulation	Real Platform
RF carrier frequency f_0	78 GHz	78 GHz
Sweep bandwidth B	1 GHz	3.5 GHz
Modulation rate K	1 GHz/μs	50 MHz/μs
Pulse repetition interval (PRI) T_s	70 μs	70 μs
Fast time sampling frequency f_s	1.2 GHz	8 MHz
Number of fast time sampling N	512	512

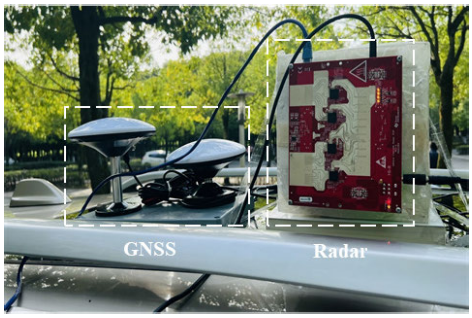


Fig. 3. Auto-SAR system platform.

A. Simulations

In the Auto-SAR simulations, first, we assume that the length of the vehicular synthetic aperture is 1 m, and to avoid the azimuth aliasing, the distance of the synthetic aperture antenna is $d = \lambda/2$. Then, we carried out two

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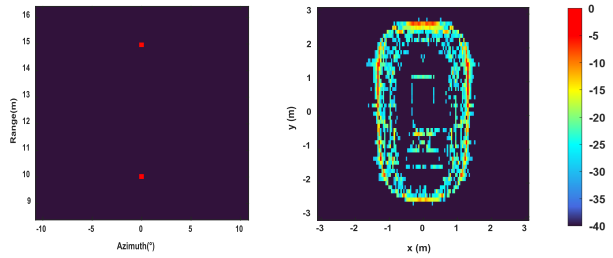


Fig. 4. Simulation models. (a) Point targets. (b) Car target.

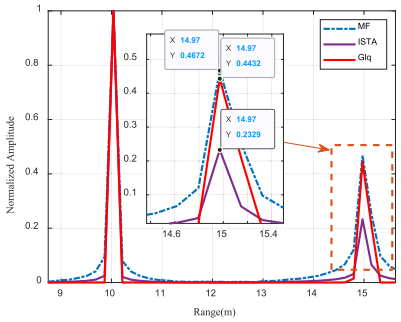


Fig. 5. Point target results of MF, ISTA, and Glq in angle slice.

simulation experiments: point targets and car targets in Fig. 4. In this simulation, the dimensions of the 2-D echo data $\mathbf{Y}$ are $N \times M$:1024 $\times$ 520, and the dimensions of 2-D imaging scene $\mathbf{X}$ are $N \times Q$:1024 $\times$ 1024.

1) Point Targets: In the simulations, two point targets with the same angle are in different range units, and their polar coordinates are (10 m, 0°) and (15 m, 0°). In order to be able to evaluate the bias effect effectively, the amplitudes of two point targets with the same angle are set to 1 and 0.5, respectively, where amplitude 1 represents a strong target (10 m) and amplitude 0.5 represents a weak target (15 m).

Fig. 5 shows the normalized amplitude estimation results of MF, ISTA, and Glq in the angle slice. For the strong target at 10 m, the amplitude can be well maintained. Comparing the estimation results of different methods for weak targets, we can see that the estimation results of MF and Glq are close, which are 0.46 and 0.44, respectively (see the blue dashed line and the red solid line). It means that Glq has great amplitude preservation, especially for weak targets. However, in the result of ISTA, the estimated amplitude is 0.23. Compared with the MF estimation, the amplitude is reduced by half, which validates effectively the bias estimation problem for ISTA based on l_1 regularization. For the Auto-SAR scene, this bias effect will seriously affect the imaging quality of weak targets and may even lose the imaging results of weak targets. Meanwhile, the experimental results of this amplitude estimation are consistent with the theoretical analysis in Section III-B, which further illustrates that the bias effect brought by l_1 regularization can be effectively suppressed by Glq regularization. Therefore, our proposed Glq can effectively avoid the influence of biased estimation, especially for weak targets.

2) Car Target: In this simulation, a car model is set to 10 m away from the Auto-SAR platform with different amplitude as shown in Fig. 4(b), and after generating the radar echoes, different SNR levels from −5 to 5 dB are added to the echo data to analyze the performance of different methods. Fig. 6 shows the imaging results of different methods at different SNR levels. Rows are represented as different methods and columns as different SNRs. Next, we will visually and numerically analyze the imaging results of the entire target.

From the visual evaluation, first, from the sidelobes and background noise suppression ability, the imaging results of MF have high sidelobe and background noise, especially at low SNR at −5 dB, and the target is almost submerged by noise, resulting in inconspicuous structural features. By contrast, the ISTA, Glq, and proposed Glq-Con sparse imaging methods all have different degrees of sidelobe and noise suppression in rows 2–4 of Fig. 6. However, due to the impact of the bias effect from l_1 regularization, there is a lack of target structure in the results of ISTA method in row 2 of Fig. 6. Although the target structure improves with the increase of SNR, it is still missing seriously, especially the weak scatterers in Fig. 7(b). By contrast, the Glq method is able to improve the amplitude of the target in row 2 of Fig. 6, effectively preserving the structural properties of the target. This is all thanks to the nonconvex property and global convergence of Glq regularization. On this basis, the proposed Glq-Con imaging method can further enhance the structural characteristics of the target because the convolution regularization exploits the correlation of neighboring scatterers in row 3 of Fig. 6. Therefore, our proposed Glq-Con imaging method can not only suppress sidelobes and noise but also effectively enhance the structural characteristics of the target, which can be better applied to ADAS. Besides, we also present the azimuth slices of the different imaging results in Fig. 7. There is a lot of noise and sidelobes in MF, and ISTA, Glq, and Glq-Con sparse imaging can effectively suppress noise and sidelobes. However, the weak scatterer region P1 is almost lost by ISTA in Fig. 7(b). By contrast, Glq improved the region P1 and Glq-Con further enhanced the results in Fig. 7(c) and (d). Meanwhile, we can get this conclusion more clearly in the local imaging results in Fig. 8.

From the numerical evaluation, the numerical results of different methods are shown in Table II. When the SNR is from −5 to 5 dB, the values of PBR are increased in turn by all imaging methods, which is in line with the facts. Meanwhile, the values of PBR of ISTA, Glq, and Glq-Con are greater than MF, which means that these spare imaging methods have better sidelobe and noise suppression capabilities. The values of PBR of Glq and Glq-Con are greater than ISTA, which means that the noise suppression ability of the proposed methods is better than that of ISTA.

For NIE, with the increase of SNR, the values of NIE decrease, which means that the imaging quality of all methods improves, which is in line with the fact. Meanwhile, the NIE of these sparse imaging methods is lower than that of MF, which means that they are better than MF. In these sparse imaging methods in the same SNR, the value of NIE of Glq-Con is higher than ISTA, even better than Glq. It means that our proposed Glq-Con has a better performance of target structure enhancement. Meanwhile, the values of NAT of Glq-Con are

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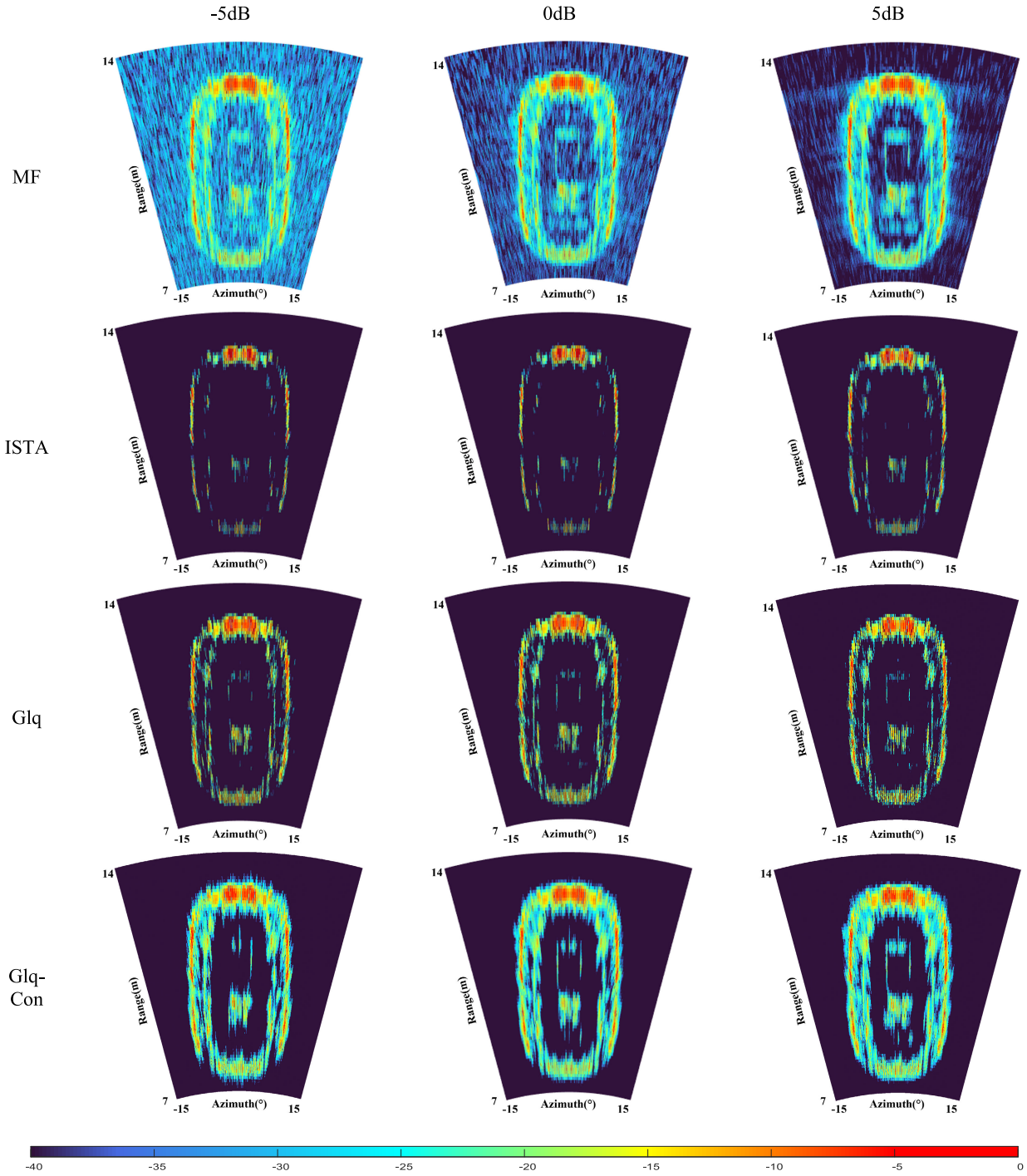


Fig. 6. Imaging results in the case of different SNRs. Row 1: MF. Row 2: ISTA. Row 3: Glq. Row 4: Glq-Con. The dynamic segmentation of imaging amplitude is 0 to −40 dB.

TABLE II

NUMERICAL COMPARISONS OF IMAGING METHOD WITH DIFFERENT SNRS, NIE, NAT, AND PBR

Methods	SNR=-5dB			SNR=0dB			SNR=5dB		
	NIE	NAT	PBR (dB)	NIE	NAT	PBR (dB)	NIE	NAT	PBR (dB)
MF	3.72	0.08	32.39	2.91	0.08	37.44	2.23	0.08	41.89
ISTA	4.41×10^{-2}	0.02	67.80	4.01×10^{-2}	0.03	70.44	3.97×10^{-2}	0.03	74.65
Glq	6.97×10^{-2}	0.04	75.16	6.75×10^{-2}	0.05	76.47	5.98×10^{-2}	0.05	79.13
Glq-Con	16.54×10^{-2}	0.08	74.86	15.40×10^{-2}	0.09	76.25	14.98×10^{-2}	0.09	79.28

better than that of Glq and ISTA and close to that of MF, which means that the structure of the target is effectively enhanced by Glq-Con. It is due to the nonconvex property and global convergence of Glq regularization, and the convolution

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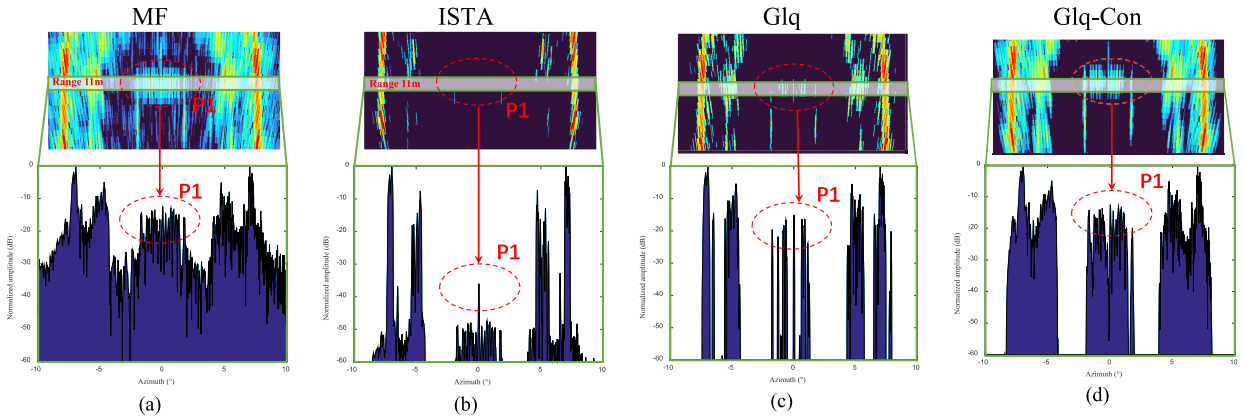


Fig. 7. Azimuth slices of the different imaging results at the same range of 11 m.

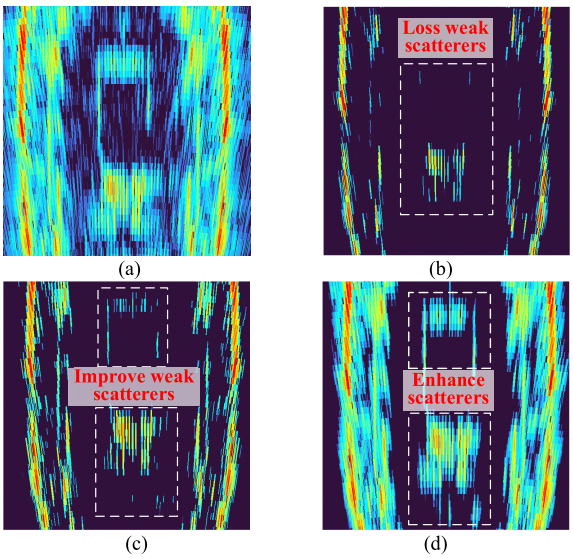


Fig. 8. Local imaging results of (a) MF, (b) ISTA, (c) Glq, and (d) Glq-Con.

regularization by exploiting the correlation of neighboring scatterers.

Therefore, from the visual evaluation and numerical evaluation, our proposed Glq-Con can not only suppress sidelobes and noise but also effectively enhance the structural characteristics of the target.

B. Experiment

In the Auto-SAR experiments, we conduct two driving scenario experiments, i.e., parking lot, and crossroad (row 1 of Fig. 9) to verify the effectiveness of the proposed imaging method, where the car targets are detected by the radar at different angles. In addition, considering the radar beamwidth, imaging resolution, and other factors, we synthesized 2-m apertures with $d = v_s \times T_s \leq \lambda/2$, and the vehicle platform v_s is 5 km/s. Then, the experimental scenarios are as follows.

1) *Parking Lot Scenario:* The scenario contains two cars parked side by side at a distance of about 8 m and the background targets (one barrier and two greenbelts), where the interval of two cars is about 1 m, and all the background targets are in the back of the car in row 1 and column 1 of Fig. 9. In this experiment, the dimensions of the 2-D echo data $\mathbf{Y}$ are $N \times M$:1856 $\times$ 1029, and the dimensions of 2-D imaging scene $\mathbf{X}$ are $N \times Q$:1856 $\times$ 1024.
2) *Crossroad Scenario:* In the second scenario, there are two cars at the crossroad; in addition, there is a tree and a streetlight beside the cars. The interval of two cars is about 0.5 m, as shown in row 1 and column 2 of Fig. 9. In this experiment, the dimensions of the 2-D echo data $\mathbf{Y}$ are $N \times M$:1748 $\times$ 1029, and the dimensions of 2-D imaging scene $\mathbf{X}$ are $N \times Q$:1748 $\times$ 1024.

TABLE III

NUMERICAL COMPARISONS OF IMAGING METHODS WITH THE PARKING LOT AND CROSSROAD SCENARIO, NIE, NAT, AND PBR

Methods	Parking lot			Crossroad		
	NIE	NAT	PBR (dB)	NIE	NAT	PBR (dB)
MF	2.08	0.05	45.86	1.80	0.04	48.96
ISTA	0.36	0.03	81.83	0.24	0.02	84.47
Glq	0.40	0.04	90.29	0.27	0.03	90.35
Glq-Con	**0.95**	**0.05**	90.43	**0.65**	**0.05**	90.28

Fig. 9 shows the imaging results of two scenarios by MF, ISTA, Glq, and our proposed Glq-Con. Fig. 10 shows the local imaging results in scenario 1. Fig. 11 shows that the azimuth slices of the different imaging results in scenario 2. Table III shows the numerical comparison of imaging methods under different scenes, NIE and PBR.

First, from the visual evaluation, the imaging results of MF have a lot of background noise and sidelobe interference. Especially, in the parking lot scenario, the target is almost drowned in background noise, and cars 1 and 2 cannot be effectively distinguished. In the crossroad scenario, there is obvious sidelobe interference in the imaging results of the streetlight in row 2 and column 2 of Fig. 9. Based on the sparse imaging model, ISTA, Glq, and Glq-Con have great noise and sidelobe suppression ability in rows 3–5 of Fig. 9. However, because of the bias effect, ISTA has a serious loss of accuracy for the imaging results of weak targets. Especially, in the imaging results in a parking lot in Fig. 10(b) and (f), it can be seen that the structure of car 1 and the greenbelt is completely lost of ISTA, but our Glq-Con effectively retains the structure of the greenbelt. Meanwhile, the Glq has better

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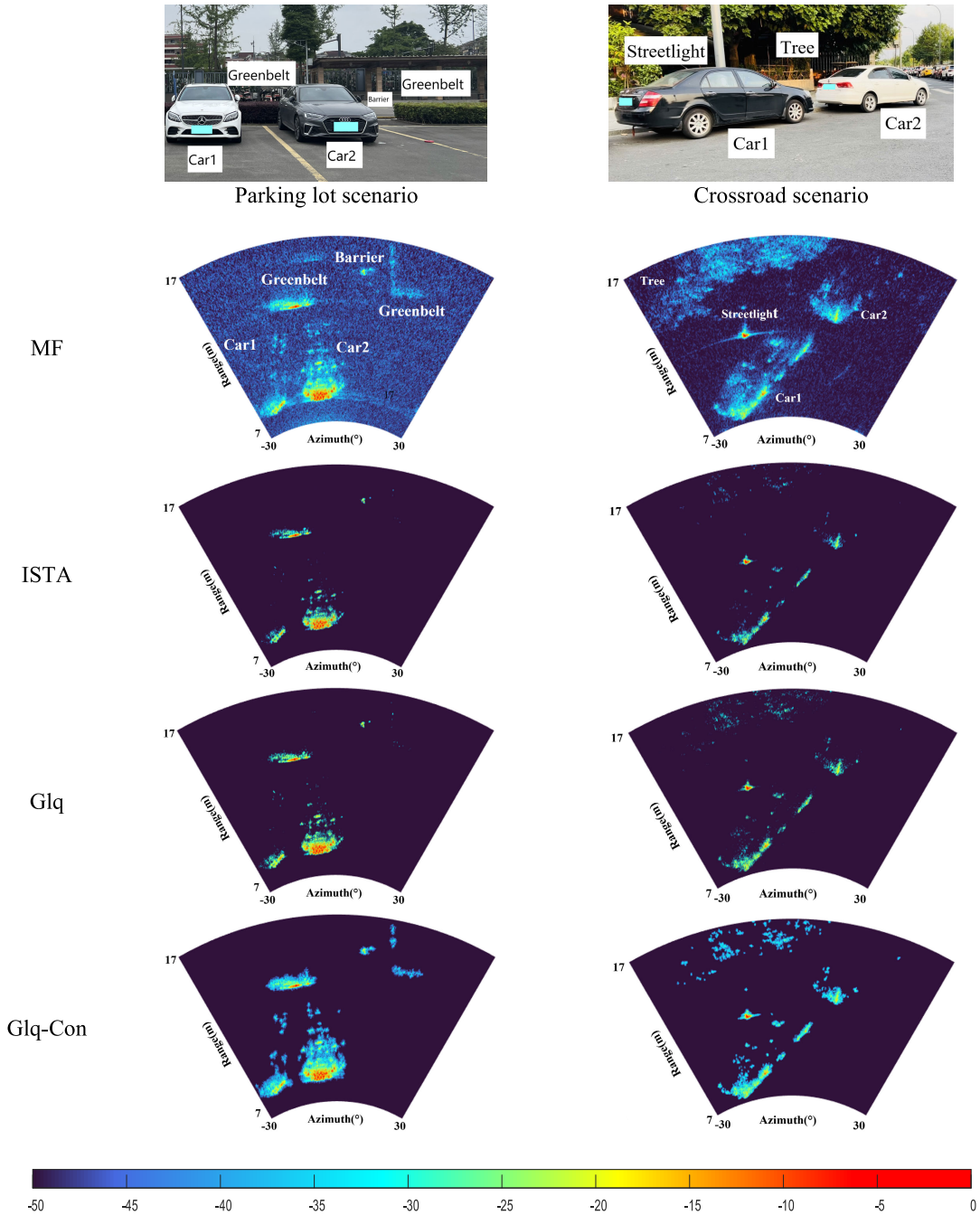


Fig. 9. Imaging results in the case of different driving scenarios. Row 1: optical scenarios. Row 2: MF. Row 3: ISTA. Row 4: Glq. Row 5: Glq-Con. Column 1: parking lot scenario. Column 2: crossroad scenario. The dynamic segmentation of imaging amplitude is 0 to −50 dB.

retention of the car contour, which can be seen in Fig. 10(c) and (g). It uses the nonconvex property and global convergence of Glq regularization to improve the amplitude of the target. Then, based on Glq, our proposed Glq-Con further enhances the amplitude of the target, which makes the target structure characteristics very prominent. In particular, the structure of the greenbelt is very clear. Besides, the slice imaging results of the streetlight in Fig. 11 show that the proposed Glq-Con has better sidelobe and noise inhibition and better retention of target amplitude than ISTA. Therefore, our proposed Glq-Con method has the best accuracy of image reconstruction and effectively enhances the structural characteristics of the target.

From the numerical evaluation in Table III, the optimal values are marked in bold. It can see the values of PBR in two scenarios, and these sparse imaging methods have a greater noise and sidelobe suppression ability than MF. The noise and sidelobe suppression ability of Glq and Glq-Con is better than that of ISTA. In the values of NIE, due to the nonconvex property and global convergence of Glq regularization, and the convolution regularization by exploiting the correlation of neighboring scatterers, our proposed Glq-Con method has better target structure retention than Glq, even than ISTA. It is improved about by 0.59 to ISTA and 0.55 to Glq in the parking lot and improved about by 0.41 to ISTA and 0.38 to Glq. In the values of NAT, our Glq-Con is better

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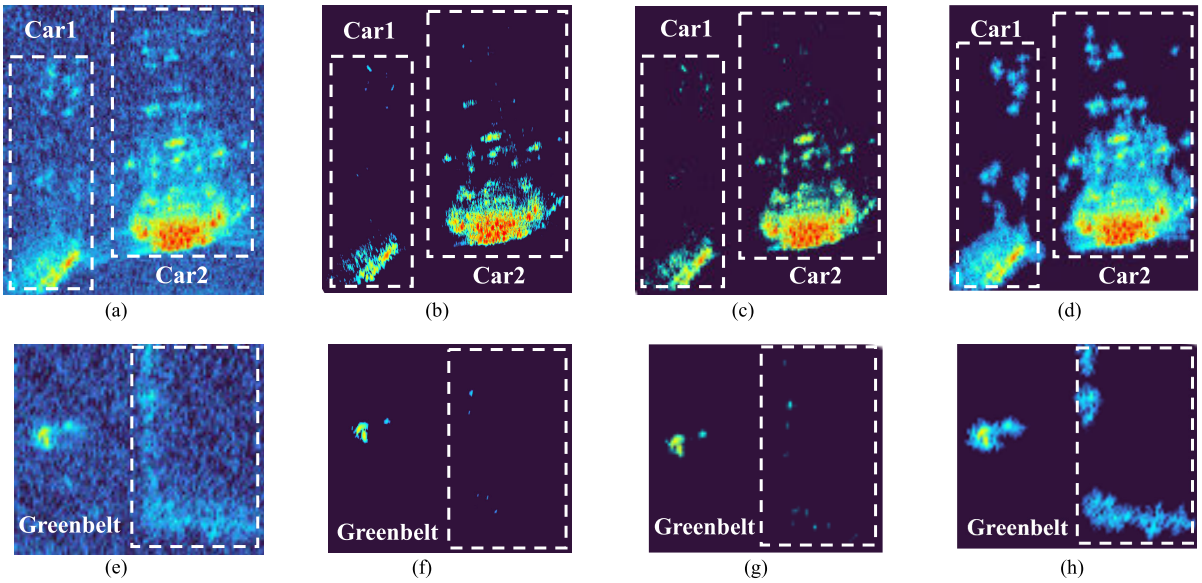


Fig. 10. Local imaging results of car and greenbelt of (a) and (e) MF, (b) and (f) ISTA, (c) and (g) Glq, and (d) and (h) Glq-Con in parking lot.

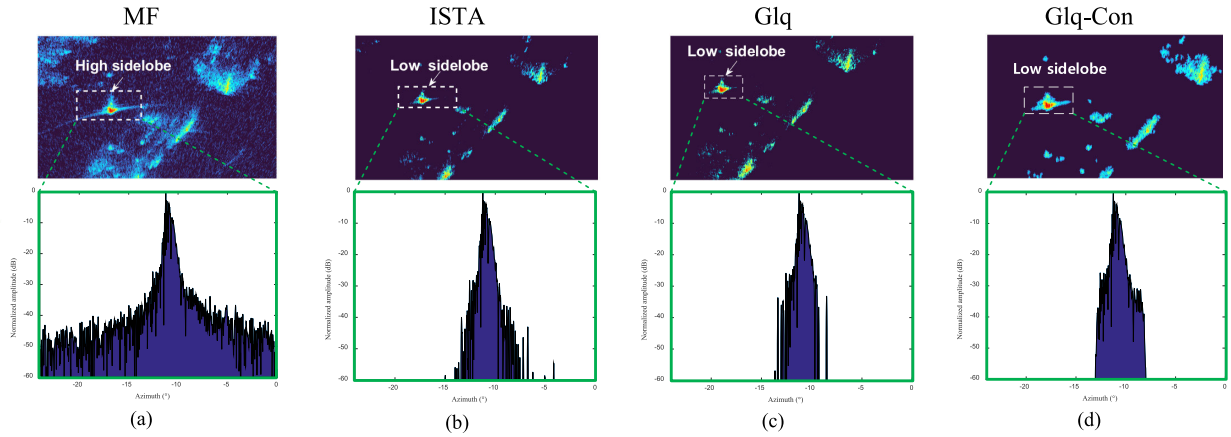


Fig. 11. Azimuth slices of streetlight of (a) MF, (b) ISTA, (c) Glq, and (d) Glq-Con in crossroad.

than Glq and ISTA in the two scenarios. Therefore, it can be considered that the proposed method enhances the structure of the target. Therefore, from the numerical evaluation, the image quality obtained by the proposed imaging method in different scenarios is better. Overall, the image quality obtained by the proposed Glq-Con imaging method is the best compared to other imaging methods, whether from visual evaluation or numerical evaluation.

C. Computation Complexity

In this section, we qualitatively and quantitatively analyze the computational complexity of different methods. Table IV shows the qualitatively computational complexity of different methods and Tables V and VI show the corresponding time cost in the parking lot and crossroad. In Table IV, it can be seen that the computational complexity is mainly dominated by the dimensions of the matrix $\mathbf{A}$ and $\mathbf{X}$ before the FFT operator. This is because matrix operations between $\mathbf{A}$ and $\mathbf{X}$ are mainly performed $\mathcal{O}(\mathrm{MNQ})$ during the imaging process. After the FFT operator, the matrix operations between $\mathbf{A}$ and $\mathbf{X}$ is replaced by the FFT operator, which leads to a sharp reduction in computational complexity $\mathcal{O}(\mathrm{MNQ}) \rightarrow \mathcal{O}(Q \log_2 Q)$, especially when the matrix dimension and the number of iterations $I_{\max}$ are large. Meanwhile, compared with ISTA, the computational complexity of Glq increases by $\mathcal{O}(\mathrm{NQ})$ and is much smaller than $\mathcal{O}(\mathrm{MNQ})$. Our proposed Glq-Con is compared with Glq, and the computational complexity of Glq increases by $\mathcal{O}(K^2\mathrm{NQ})$, which is much smaller than $\mathcal{O}(\mathrm{MNQ})$ too. It means that Glq and our proposed Glq-Con can improve the image quality without adding too much computational burden. Therefore, our proposed imaging method not only improves the imaging quality but also has low computational complexity, which provides great potential for real-time or semi-real-time imaging of Auto-SAR in the future.

Tables V and VI show the time cost in the parking lot and crossroad. It should be noted that since in the experiment, the matrix dimension we set is not very large, and considering that MTALAB has a certain acceleration processing for matrix operations. Thus, although our proposed FFT operator-based method can reduce the running time by about 1 s, this improvement is still significant. At the same time, with the increase of the matrix dimension and the number of iterations, such a gap will be gradually obvious. Therefore, based on our proposed efficient imaging model, the computational complexity is effectively reduced.

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TABLE IV
COMPUTATION COMPLEXITY OF IMAGING METHODS

Methods	Computation complexity	Method in FFT operator	Computation complexity
MF	$\mathcal{O}(MNQ)$	MF	$\mathcal{O}(Q\log_2 Q)$
ISTA	$\mathcal{O}\{I_{max}(2MNQ)\}$	ISTA	$\mathcal{O}\{I_{max}(2Q\log_2 Q)\}$
Glq	$\mathcal{O}\{I_{max}(2MNQ+2NQ)\}$	Glq	$\mathcal{O}\{I_{max}(2Q\log_2 Q+2NQ)\}$
Glq-Con	$\mathcal{O}\{I_{max}(2MNQ+2NQ+K^2NQ)\}$	Glq-Con	$\mathcal{O}\{I_{max}(2Q\log_2 Q+2NQ+K^2NQ)\}$

TABLE V
TIME COST OF IMAGING METHODS IN PARKING LOT

Methods	Time cost	Method via FFT	Time cost
MF	2.82×10^{-2}s	MF	9.33×10^{-3}s
ISTA	4.88s	ISTA	4.09s
Glq	6.01s	Glq	5.07s
Glq-Con	6.20s	Glq-APC	5.28s

TABLE VI
TIME COST OF IMAGING METHODS IN CROSSROAD

Methods	Time cost	Methods via FFT	Time cost
MF	2.63×10^{-2}s	MF	9.53×10^{-3}s
ISTA	4.19s	ISTA	3.32s
Glq	4.50s	Glq	3.89s
Glq-Con	4.61s	Glq-APC	3.92s

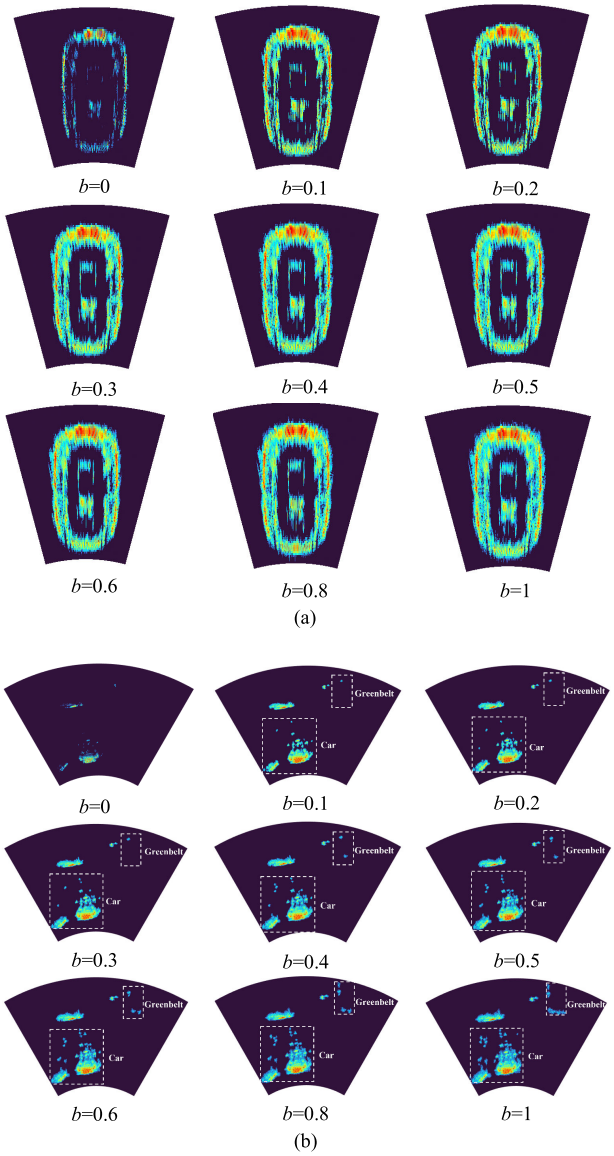


Fig. 12. Results of the simulation and experiment at different values of b. (a) Simulation results at different values of b. (b) Experiment results of at different values of b in parking lot.

D. Discussion of the Parameter b and Motion Errors

In this section, we primarily discuss the impact of parameter b selection and motion errors on imaging results.

1) Parameter b in the Convolution Kernel **C***:* In convolutional regularization, the convolution kernel $\mathbf{C}$ is used to enhance the structure of goal continuity, while the value of b affects the enhancement performance of $\mathbf{C}$. Considering that the value of the scattering center point is a, thus, the range of values of b can be considered [0, a]. Next, we present the results of the simulation and experiment at different values of b in Fig. 12, where a is set to 1.

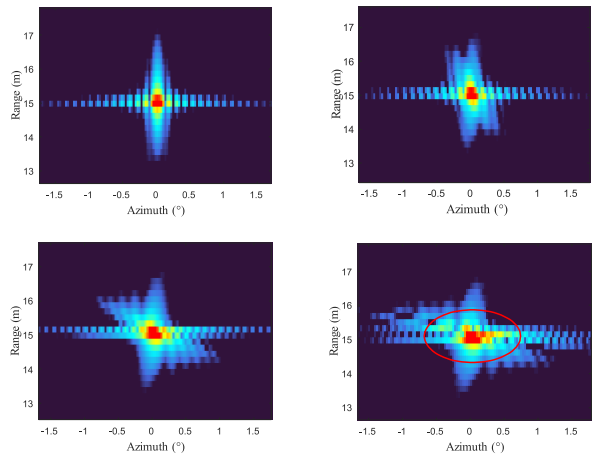
From the visual evaluation, first, in the simulation results, when $b \geq 0.5$, the enhancement performance of the proposed Glq-Con is pretty close and great; when $0 < b < 0.5$, the enhancement performance of the proposed Glq-Con is relatively weakened; and when $b = 0$, Glq-Con cannot enhance the imaging results. Then, in the experiment results, for the cars, when $b \geq 0.5$, the structure of cars is well enhanced by Glq-Con, and for the greenbelt, when $b = 1$, the greenbelt has the best imaging results.

Therefore, when the imaging target is a strong scatterer, such as car, $b \in [0.5, a]$ all can have great imaging results, and when the imaging scene contains weak scatterer targets, such as greenbelt, $b = 1$ provides better reconstruction of the structure.

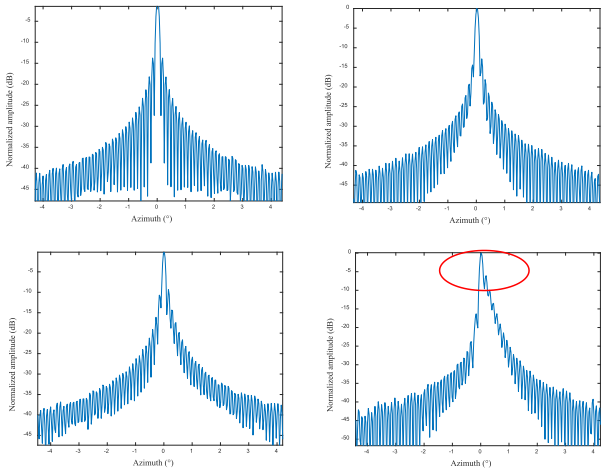
2) Platform Motion Error: To analyze the effect of platform motion errors on imaging results, we take the point target as the simulation experiment and use the uniform acceleration motion model to represent the platform motion model. Assuming that the initial velocity of the vehicle is $v_s = 5$ km/s, the acceleration is β, and the synthetic aperture length is 1 m by a uniform velocity of 5 km/s about 0.7 s. Fig. 13 shows the imaging results of point targets at different accelerations β.

In Fig. 13(a), when the platform is ideally moving at a uniform velocity, i.e., $\beta = 0$, point targets can be well focused,

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(a)



(b)

Fig. 13. Platform motion error simulation. (a) Imaging results: 0, 2, 5, and 8. (b) Azimuth slice in range 15 m.

and when the change in platform velocity is small (β = 2,5), the point target can still be focused. However, when the velocity of the vehicle changes greatly (β = 8), the point target has a certain defocus. In the azimuth slice results in Fig. 13(b), as β keeps getting larger, the sidelobes of the point target are deformed to different degrees. However, when β is larger, the point target cannot be well focused. Therefore, when the platform motion errors are small, the imaging results are relatively well focused. However, when the platform motion errors are significant, we need to compensate for these errors, such as using autofocus algorithms, to achieve well-focused imaging results.

Therefore, in order to avoid the movement error of the vehicle platform as much as possible, we can reasonably set a smaller aperture length of Auto-SAR (about 1∼2 m). When the velocity of the vehicle platform is 5 km/s, it only takes about 1.4 s to synthesize a 2-m aperture, and when the velocity of the car platform is 10 km/s, even it only takes about 0.7 s. Therefore, during this short period of time, it is reasonable to assume that the platform is moving at a constant velocity. At this time, Auto-SAR, for a target at 10 m, can provide an azimuth resolution of about 0.01 m. Then, for the distance of the Auto-SAR antenna, considering that the maximum beamwidth of mmW automotive radar is $-60°$ to $60°$ and to avoid the azimuth aliasing, we set the array element distance of Auto-SAR to be $d = v_s \times T_s \leq \lambda/2$. Therefore, we can reasonably set the pulse repetition interval (PRI) T_s of the transmitted signal to satisfy this condition. In our experiment, the vehicle platform is 5 km/s, so we can set to $T_s = 70\mu s$ to satisfy this condition. Finally, to obtain a high range resolution, we set the sweep bandwidth of the transmitted signal to 3.5 GHz.

V. Conclusion

In this article, we propose a Glq-Con regularization to enhance mmW Auto-SAR sparse imaging. Specifically, a Glq regularization is used to reduce the bias effect of l_1 regularization in order to improve the target amplitude. Then, a Con regularization to modify the reconstructed amplitude to improve the target structure. Finally, an efficient sparse imaging model is established. In this model, an FFT operator is used to replace matrix operation to reduce computational complexity, and an efficient optimizer is used to solve the imaging mode. The proposed method improves the reconstruction accuracy of images compared with imaging methods based on MF, such as sidelobe suppression. Meanwhile, the proposed method further enhances the structural features of the target compared with imaging methods based on l_1 regularization. Both simulation and experiments verify that the proposed method has a significant effect on improving the Auto-SAR image quality.

In future work, based on our vehicle platform, we will continue to focus on exploring the self-focusing Auto-SAR sparse imaging method to reduce the error of the motion trajectory to further improve the imaging quality. Also, we will focus on exploring high-quality imaging methods under arbitrary motion trajectories to cater for a variety of different driving environments.

Appendix
The Derivation of (14)

When x satisfies the conditions of (13), $f(x)$ becomes a convex function and a concave function, so we can set a generalized soft thresholding $\beta^{\mathrm{Glq}}(\lambda)$ to solve these problems. When $y < \beta^{\mathrm{Glq}}(\lambda)$, $x = 0$ is the global minimum in the range of $(0, x_0)$ in $f(x)$. Meanwhile, to guarantee that $f(x)$ has a minimum in $(x_0, +\infty)$, we should further require

$$f'(x^*) \leq 0. \tag{34}$$

The above conditions make x^* it a unique solution in $(x_0, +\infty)$ when $y > \beta^{\mathrm{Glq}}(\lambda)$. In order for x^* to be a globally optimal solution, we should also further require

$$f(x^*) \leq f(0). \tag{35}$$

Therefore, considering the boundary cases of the above two conditions (34) and (35) and bringing them into (10) and (11), we can obtain the equality in (13) as follows:

$$\begin{cases} \{x^* - \beta^{\mathrm{Glq}}(\lambda)\}^2 + 2\lambda (x^*)^q = \{\beta^{\mathrm{Glq}}(\lambda)\}^2 \\ x^* - \beta^{\mathrm{Glq}}(\lambda) + \lambda q (x^*)^{q-1} = 0. \end{cases} \tag{36}$$

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REFERENCES

[1] C. Waldschmidt, J. Hasch, and W. Menzel, "Automotive radar—From first efforts to future systems," *IEEE J. Microw.*, vol. 1, no. 1, pp. 135–148, Jan. 2021.

[2] S. M. Patole, M. Torlak, D. Wang, and M. Ali, "Automotive radars: A review of signal processing techniques," *IEEE Signal Process. Mag.*, vol. 34, no. 2, pp. 22–35, Mar. 2017.

[3] F. Engels, P. Heidenreich, A. M. Zoubir, F. K. Jondral, and M. Wintermantel, "Advances in automotive radar: A framework on computationally efficient high-resolution frequency estimation," *IEEE Signal Process. Mag.*, vol. 34, no. 2, pp. 36–46, Mar. 2017.

[4] H. Yan et al., "Overview of signal processing techniques for automotive millimeter-wave radar," *J. Radars*, vol. 12, no. 5, pp. 923–970, 2023.

[5] J. Hasch, E. Topak, R. Schnabel, T. Zwick, R. Weigel, and C. Waldschmidt, "Millimeter-wave technology for automotive radar sensors in the 77 GHz frequency band," *IEEE Trans. Microw. Theory Techn.*, vol. 60, no. 3, pp. 845–860, Mar. 2012.

[6] Y. Xu, X. Zhang, S. Wei, J. Shi, T. Zeng, and T. Zhang, "A target-oriented Bayesian compressive sensing imaging method with region-adaptive extractor for mmW automotive radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5213716.

[7] J. M. Merlo and J. A. Nanzer, "A C-band fully polarimetric automotive synthetic aperture radar," *IEEE Trans. Veh. Technol.*, vol. 71, no. 3, pp. 2587–2600, Mar. 2022.

[8] T. Grebner, R. Riekenbrauck, and C. Waldschmidt, "Simultaneous localization and mapping (SLAM) for synthetic aperture radar (SAR) processing in the field of autonomous driving," *IEEE Trans. Radar Syst.*, vol. 2, pp. 47–66, 2024.

[9] M. Steiner, T. Grebner, and C. Waldschmidt, "Millimeter-wave SAR-imaging with radar networks based on radar self-localization," *IEEE Trans. Microw. Theory Techn.*, vol. 68, no. 11, pp. 4652–4661, Nov. 2020.

[10] X. Gao, S. Roy, and G. Xing, "MIMO-SAR: A hierarchical high-resolution imaging algorithm for mmWave FMCW radar in autonomous driving," *IEEE Trans. Veh. Technol.*, vol. 70, no. 8, pp. 7322–7334, Aug. 2021.

[11] T. Grebner, A. Grathwohl, P. Schoeder, V. Janoudi, and C. Waldschmidt, "Probabilistic SAR processing for high-resolution mapping using millimeter-wave radar sensors," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 5, pp. 4800–4814, 2023.

[12] S. Tebaldini et al., "Sensing the urban environment by automotive SAR imaging: Potentials and challenges," *Remote Sens.*, vol. 14, no. 15, p. 3602, Jul. 2022.

[13] M. Manzoni, S. Tebaldini, A. V. Monti-Guarnieri, C. M. Prati, and I. Russo, "A comparison of processing schemes for automotive MIMO SAR imaging," *Remote Sens.*, vol. 14, no. 19, p. 4696, Sep. 2022.

[14] A. Laribi, M. Hahn, J. Dickmann, and C. Waldschmidt, "Performance investigation of automotive SAR imaging," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Apr. 2018, pp. 1–4.

[15] B. Zhang, G. Xu, R. Zhou, H. Zhang, and W. Hong, "Multi-channel back-projection algorithm for Mmwave automotive MIMO SAR imaging with Doppler-division multiplexing," *IEEE J. Sel. Topics Signal Process.*, vol. 17, no. 2, pp. 445–457, Mar. 2023.

[16] S. Mehrdad, *Synthetic Aperture Radar Signal Processing*. New York, NY, USA: Wiley, 1999.

[17] E. Jansing, *Introduction to Synthetic Aperture Radar: Concepts and Practice*. New York, NY, USA: McGraw-Hill, 2021.

[18] Y. C. Eldar and G. Kutyniok, *Compressed Sensing: Theory and Applications*. Cambridge, U.K.: Cambridge Univ. Press, 2012.

[19] D. L. Donoho, "Compressed sensing," *IEEE Trans. Inf. Theory*, vol. 52, no. 4, pp. 1289–1306, Apr. 2006.

[20] M. Wang, S. Wei, Z. Zhou, J. Shi, and X. Zhang, "Efficient ADMM framework based on functional measurement model for mmW 3-D SAR imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5226417.

[21] M. Wang, S. Wei, Z. Zhou, J. Shi, X. Zhang, and Y. Guo, "CTV-net: Complex-valued TV-driven network with nested topology for 3-D SAR imaging," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 4, pp. 5588–5602, Apr. 2024.

[22] M. Wang, S. Wei, J. Shi, X. Zhang, and Y. Guo, "3-D SAR imaging via perceptual learning framework with adaptive sparse prior," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5202716.

[23] S. Wei et al., "Learning-based split unfolding framework for 3-D mmW radar sparse imaging," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5229317.

[24] S. Wei et al., "Nonline-of-sight 3-D imaging using millimeter-wave radar," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5106518.

[25] G. Xu, B. Zhang, J. Chen, F. Wu, J. Sheng, and W. Hong, "Sparse inverse synthetic aperture radar imaging using structured low-rank method," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5213712.

[26] M. Tello Alonso, P. Lopez-Dekker, and J. J. Mallorqui, "A novel strategy for radar imaging based on compressive sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 12, pp. 4285–4295, Dec. 2010.

[27] W. Pu, "SAE-net: A deep neural network for SAR autofocus," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5220714.

[28] R. Tibshirani, "Regression shrinkage and selection via the lasso," *J. Roy. Stat. Soc. B, Stat. Methodol.*, vol. 58, no. 1, pp. 267–288, Jan. 1996.

[29] A. Beck and M. Teboulle, "A fast iterative shrinkage-thresholding algorithm for linear inverse problems," *SIAM J. Imag. Sci.*, vol. 2, no. 1, pp. 183–202, Jan. 2009.

[30] D. L. Donoho, A. Maleki, and A. Montanari, "Message-passing algorithms for compressed sensing," *Proc. Nat. Acad. Sci. USA*, vol. 106, no. 45, pp. 18914–18919, Jul. 2009.

[31] B. K. Natarajan, "Sparse approximate solutions to linear systems," *SIAM J. Comput.*, vol. 24, no. 2, pp. 227–234, Apr. 1995.

[32] F. Wen, L. Chu, P. Liu, and R. C. Qiu, "A survey on nonconvex regularization-based sparse and low-rank recovery in signal processing, statistics, and machine learning," *IEEE Access*, vol. 6, pp. 69883–69906, 2018.

[33] J. Fan and R. Li, "Variable selection via nonconcave penalized likelihood and its Oracle properties," *J. Amer. Stat. Assoc.*, vol. 96, no. 456, pp. 1348–1360, 2001.

[34] T. S. Cho, N. Joshi, C. L. Zitnick, S. B. Kang, R. Szeliski, and W. T. Freeman, "A content-aware image prior," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2010, pp. 169–176.

[35] M. Lai and J. Wang, "An unconstrained ℓ_q minimization with 0q$\leq$1 for sparse solution of underdetermined linear systems," *SIAM J. Optim.*, vol. 21, no. 1, pp. 82–101, 2011.

[36] E. Candes, M. Wakin, and S. Boyd, "Enhancing sparsity by reweighted ℓ_1 minimization," *J. Fourier Anal. Appl.*, vol. 14, no. 5, pp. 877–905, 2008.

[37] W. Zuo, D. Meng, L. Zhang, X. Feng, and D. Zhang, "A generalized iterated shrinkage algorithm for non-convex sparse coding," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 217–224.

[38] Y. Xu, X. Zhang, S. Wei, S. Jun, and T. Zeng, "A fast super-resolution imaging method via subspace detection and near-field phase compensation for mmW automotive radar," in *Proc. IGARSS*, Pasadena, CA, USA, Jul. 2023, pp. 4415–4418.

[39] J. Fang, Z. Xu, B. Zhang, W. Hong, and Y. Wu, "Fast compressed sensing SAR imaging based on approximated observation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 1, pp. 352–363, Jan. 2014.

[40] S. Osher, F. Ruan, J. Xiong, Y. Yao, and W. Yin, "Sparse recovery via differential inclusions," *Appl. Comput. Harmon. Anal.*, vol. 41, no. 2, pp. 436–469, Sep. 2016.

[41] Z. Xu, X. Chang, F. Xu, and H. Zhang, "$L_{1/2}$ regularization: A thresholding representation theory and a fast solver," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 23, no. 7, pp. 1013–1027, Jul. 2012.

[42] S. Wei et al., "3DRIED: A high-resolution 3-D millimeter-wave radar dataset dedicated to imaging and evaluation," *Remote Sens.*, vol. 13, no. 17, p. 3366, 2021.

[43] B. Hou, Z. Wen, L. Jiao, and Q. Wu, "Target-oriented high-resolution SAR image formation via semantic information guided regularizations," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1922–1939, Apr. 2018.

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